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CUET UG

Physics (322)

Electronic Devices

Official Syllabus Unit 9 · Class 12 NCERT · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 9)	Key Exam Focus & Skills
Energy Bands in Solids	Energy bands in conductors, semiconductors and insulators (qualitative ideas only).	Valence Band, Conduction Band, forbidden energy gap E_g; classification criteria: Insulators ($E_g > 3$ eV), Semiconductors ($0 < E_g < 3$ eV), Metals ($E_g = 0$ or overlapping); temperature dependence of conductivity ($\alpha < 0$).
Intrinsic & Extrinsic Semiconductors	Intrinsic and extrinsic semiconductors, p-type and n-type.	Elemental (Si, Ge) and compound semiconductors; thermal pair generation ($n_e = n_h = n_i$); pentavalent donor doping (n-type: $n_e \gg n_h$); trivalent acceptor doping (p-type: $n_h \gg n_e$); mass-action law $n_e n_h = n_i^2$; electrical neutrality.
p-n Junction & Diode Characteristics	p-n junction Semiconductor diode its I-V characteristics in forward and reverse bias.	Diffusion and drift currents; space-charge depletion layer and barrier potential V_0 (0.7 V for Si, 0.3 V for Ge); forward bias barrier lowering and knee voltage; reverse bias saturation current I_s; dynamic resistance $r_d = \Delta\frac{V}{\Delta}I$.
Diode as a Rectifier	Application of junction diode - diode as a rectifier.	Half-wave rectifier ($f_{\text{out}} = f_{\text{in}}$, $\eta_{\max} = 40.6\%$, ripple $r = 1.21$); full-wave center-tapped and bridge rectifiers ($f_{\text{out}} = 2f_{\text{in}}$, $\eta_{\max} = 81.2\%$, ripple $r = 0.48$); smoothing capacitor filter action.
Special Purpose Diodes	Zener diode and its characteristics, voltage regulator; optoelectronic devices (photodiode, LED, solar cell).	Zener breakdown mechanism (high field, thin depletion); constant voltage shunt regulator design ($R_S = \frac{V_{\text{in}} - V_Z}{I_S}$); photodiode reverse-bias minority detec-

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 9)	Key Exam Focus & Skills
		tion; LED forward-bias radiative recombination; solar cell fourth-quadrant V_{oc} and I_{sc}.
Circuits, Traps & Master Cheatsheet	Synthesis, circuit analysis and quantitative problem solving.	Ideal diode switching rules; conducting vs reverse cut-off logic; rectifier frequency conversions; Zener current partitioning ($I_S = I_Z + I_L$); comprehensive formula matrix.

EXAM PATTERN

CUET UG Exam Architecture: Unit 9 (Electronic Devices) consistently yields 3 to 5 questions in Section II Physics. High-frequency question archetypes include: (1) Energy band gap thresholds (E_g of Si, Ge, Diamond); (2) Mass-action law calculations ($n_e n_h = n_i^2$); (3) Depletion layer behavior under forward/reverse bias; (4) Rectifier output ripple frequency (50 Hz $\rightarrow$ 50 Hz vs 100 Hz); and (5) Zener diode series resistance sizing (R_S).

Concept Module 1: Energy Bands in Solids & Material Classification

Band Formation in Crystalline Solids

In an isolated, single atom, electrons occupy well-defined, discrete energy levels governed by quantum mechanics. However, when N atoms assemble to form a crystalline solid (such as Silicon or Germanium with interatomic spacing $d \approx 2$ to 3 Å), the outer valence electron shells overlap strongly.

- Due to the Pauli exclusion principle, each discrete atomic energy level splits into N exceedingly closely spaced sub-levels, forming a virtually continuous **Energy Band**.
- **Valence Band (VB):** The highest energy band occupied by valence electrons at absolute zero (0 K). It may be completely filled or partially filled, but never completely empty.
- **Conduction Band (CB):** The lowest unoccupied or partially filled energy band above the valence band. Electrons residing in the conduction band are free to drift under an external electric field.
- **Forbidden Energy Gap (E_g):** The energy separation between the top of the valence band (E_v) and the bottom of the conduction band (E_c). No stable quantum electron states exist within this gap:
$$E_g = E_c - E_v$$

KEY FORMULA

Energy Band Classification Criteria:

Insulator: $E_g > 3$ eV (e.g., Diamond: $E_g \approx 5.4$ eV to 6 eV)

Semiconductor: $0 < E_g < 3$ eV (Silicon: 1.1 eV, Germanium: 0.7 eV, GaAs: 1.43 eV)

Conductor / Metal: $E_g = 0$ (Conduction & Valence bands overlap, e.g., Na, Mg, Cu)

Classification of Conductors, Semiconductors & Insulators

1. **Metals (Conductors):**
   - Either the valence band and conduction band overlap (as in Bivalent metals like Zinc and Magnesium), or the conduction band is partially filled (as in Monovalent metals like Sodium and Copper).
   - Millions of free electrons are readily available at all temperatures. Resistivity is low ($\rho \sim 10^{-8}$ to 10^{-2} $\Omega \cdot \text{m}$).
   - **Temperature Effect:** As temperature increases, lattice vibrations increase electron scattering, decreasing relaxation time τ. Thus, resistivity increases ($\alpha > 0$, positive temperature coefficient of resistance).
2. **Insulators:**
   - The valence band is completely full, the conduction band is completely empty, and the forbidden gap is wide ($E_g > 3$ eV).
   - Even at elevated room temperatures, thermal energy ($k_B T \approx 0.026$ eV) is grossly insufficient to bridge the gap.
   - Electrical conductivity is vanishingly small; resistivity is extraordinarily high ($\rho \sim 10^{11}$ to 10^{19} $\Omega \cdot \text{m}$).
3. **Semiconductors:**
   - Possess a small, finite forbidden energy gap ($E_g < 3$ eV).
   - At absolute zero ($T = 0$ K), the valence band is completely full and the conduction band is entirely empty. A pure semiconductor behaves as a **perfect insulator** at 0 K.
   - At room temperature ($T \approx 300$ K), thermal agitation imparts enough energy to a small fraction of valence electrons to overcome E_g, jumping into the conduction band and leaving behind vacant states (**holes**) in the valence band.
   - Resistivity is intermediate ($\rho \sim 10^{-5}$ to 10^{6} $\Omega \cdot \text{m}$).
   - **Temperature Effect:** Increasing temperature excites exponentially more carriers across E_g, dramatically reducing resistivity ($\alpha < 0$, negative temperature coefficient of resistance).

IMPORTANT CONCEPT

CUET High-Yield Distinction: Carbon (Diamond), Silicon (Si), and Germanium (Ge) all belong to Group 14 and have 4 valence electrons. Yet:

- Carbon (Diamond) has $E_g \approx 5.4$ eV $\Longrightarrow$ **Insulator** (valence electrons are tightly bound close to the carbon nucleus).
- Silicon has $E_g = 1.1$ eV $\Longrightarrow$ **Semiconductor**.
- Germanium has $E_g = 0.7$ eV $\Longrightarrow$ **Semiconductor** (outer electrons are farthest from nucleus, easiest to liberate).

Concept Module 2: Intrinsic & Extrinsic Semiconductors

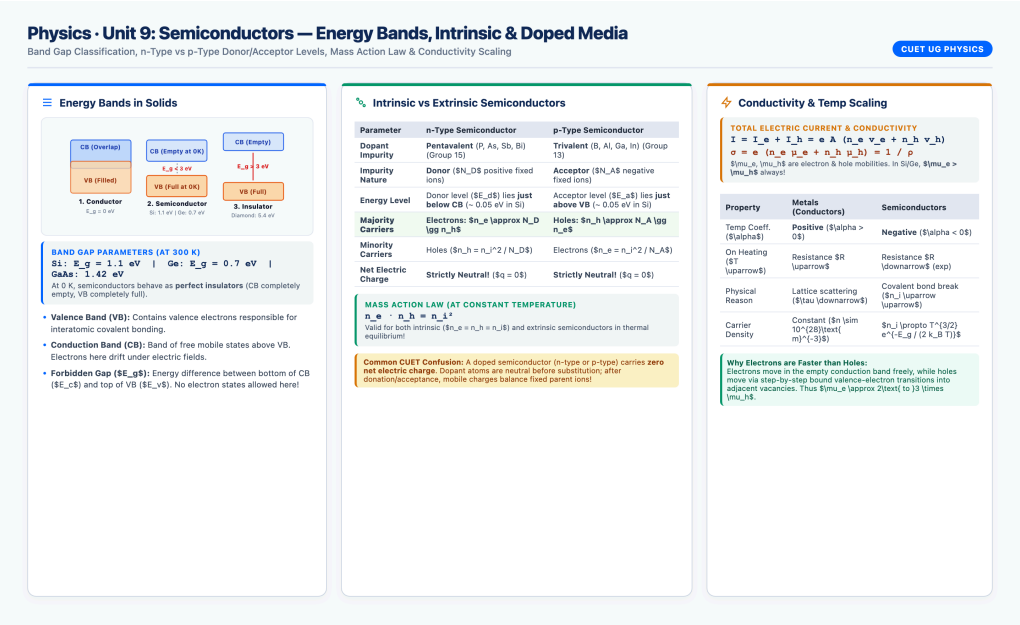


Figure 1: Energy band structures in solids, intrinsic hole-electron pair generation, and donor/acceptor energy levels in extrinsic semiconductors.

Intrinsic Semiconductors (Pure Crystals)

An intrinsic semiconductor is a chemically pure semiconductor crystal devoid of impurity atoms. Elemental Silicon ($Z = 14$) and Germanium ($Z = 32$) form a diamond-cubic tetrahedral lattice where each atom shares its 4 valence electrons with 4 adjacent neighbors via covalent bonds.

- **Thermal Generation:** At $T > 0$ K, thermal vibrations rupture occasional covalent bonds, freeing electrons into the conduction band.
- **Concept of a Hole:** The vacant electron site left behind in a covalent bond behaves as an effective positive charge $+e$ with effective mass m_h^*. An adjacent valence electron can jump into this vacancy, shifting the hole in the opposite direction.
- In an intrinsic semiconductor, free electrons and holes are generated strictly in pairs:

$$n_e = n_h = n_i$$

where n_e is the electron density in the CB, n_h is the hole density in the VB, and n_i is the **intrinsic carrier concentration**.

- **Total Conduction Current:** Both electrons in the conduction band and holes in the valence band contribute in the same direction to electric current:

$$I = I_e + I_h = eA(n_e v_e + n_h v_h)$$

$$\sigma = e(n_e \mu_e + n_h \mu_h) = e n_i (\mu_e + \mu_h)$$

where μ_e and μ_h are electron and hole mobilities ($\mu = \frac{v_d}{E}$). Note that electron mobility is always higher than hole mobility ($\mu_e > \mu_h$) because electrons move in the free conduction band, whereas holes propagate through constrained valence bond hopping.

Extrinsic Semiconductors & Doping

To enhance electrical conductivity for device applications, controlled trace amounts of specific impurity atoms (1 part in 10^6) are introduced into the intrinsic semiconductor lattice—a process known as **Doping**.

1. **n-Type Semiconductor (Pentavalent Doping):**
   - **Dopants:** Group 15 elements with 5 valence electrons—Phosphorus (P), Arsenic (As), Antimony (Sb).
   - Four valence electrons form covalent bonds with 4 neighboring Silicon atoms; the fifth electron is weakly bound (~ 0.05 eV in Si, ~ 0.01 eV in Ge) and easily ionized into the conduction band at room temperature.
   - **Donor Level (E_D):** Located just below the conduction band edge ($\Delta E_D \approx 0.01$ to 0.05 eV).

- **Carrier Densities:** Electrons are **majority carriers**; holes are **minority carriers**:
$$n_e \approx N_D \gg n_h$$
where N_D is the donor impurity concentration.

2. **p-Type Semiconductor (Trivalent Doping):**
   - **Dopants:** Group 13 elements with 3 valence electrons—Boron (B), Aluminium (Al), Indium (In), Gallium (Ga).
   - The dopant atom forms 3 covalent bonds with neighbors, leaving an empty bond vacancy (a hole) ready to accept an electron from an adjacent bond.
   - **Acceptor Level (E_A):** Located just above the top of the valence band edge ($\Delta E_A \approx 0.01$ to 0.05 eV).
   - **Carrier Densities:** Holes are **majority carriers**; electrons are **minority carriers**:
$$n_h \approx N_A \gg n_e$$
where N_A is the acceptor impurity concentration.

HIGH-YIELD TIP

CRITICAL CUET RULE: Electrical Neutrality of Doped Crystals A common misconception is that an n-type semiconductor is negatively charged and a p-type semiconductor is positively charged. **This is completely false!**

- In n-type: each donor atom donates a negative electron, becoming a fixed, immobile positive ion (N_D^+). Total positive charge (ions + holes) equals total negative charge (electrons).
- Both n-type and p-type semiconductors are **strictly electrically neutral**.

The Mass-Action Law

In thermal equilibrium, the product of electron and hole concentrations in any semiconductor (intrinsic or extrinsic) is a constant at a given temperature:
$$n_e \cdot n_h = n_i^2$$

CARRIER DENSITY VIA MASS ACTION LAW

Problem: A pure Silicon crystal ($n_i = 1.5 \times 10^{16}$ m^{-3} at 300 K) is doped with Indium atoms at a concentration of $N_A = 4.5 \times 10^{22}$ m^{-3}. Calculate the resulting electron and hole concentrations in the doped sample.
Solution: Indium is trivalent, producing a p-type semiconductor. Assuming complete ionization at room temperature:
$$n_h \approx N_A = 4.5 \times 10^{22} \text{ m}^{-3}$$
Applying the Mass-Action Law $n_e \cdot n_h = n_i^2$:
$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2}{4.5 \times 10^{22}} = \frac{2.25 \times 10^{32}}{4.5 \times 10^{22}} = 5.0 \times 10^{9} \text{ m}^{-3}$$
Interpretation: Doping with Indium drastically enhances hole density to 4.5×10^{22} m^{-3} while suppressing electron density to 5.0×10^{9} m^{-3}.

Concept Module 3: Formation of p-n Junction & I-V Characteristics

Junction Formation & The Depletion Region

When a p-type semiconductor and an n-type semiconductor are metallurgically fused together, two simultaneous carrier transport processes emerge across the interface:

1. **Diffusion Process:** Due to the steep concentration gradient, majority holes diffuse from the p-side to the n-side, and majority electrons diffuse from the n-side to the p-side, setting up a **diffusion current** directed from p to n.

2. **Formation of Space-Charge Layer:** When electrons and holes diffuse across the junction, they recombine near the interface. This exposes unneutralized, immobile ionized donors (N_D^+) on the n-side and ionized acceptors (N_A^-) on the p-side.
3. **Depletion Layer (W):** The region flanking the interface becomes completely depleted of free, mobile charge carriers. Its typical thickness is of the order of $W \approx 0.1\ \mu\text{m} = 10^{-7}$ m to $1\ \mu\text{m} = 10^{-6}$ m.
4. **Built-in Electric Field (E_i):** The immobile ion layer creates an internal electric field directed from positive donor ions (n-side) to negative acceptor ions (p-side), opposing further diffusion.
5. **Drift Process:** The electric field sweeps minority carriers across the junction (electrons from p to n, holes from n to p), generating a **drift current** directed from n to p.
6. **Thermal Equilibrium:** Equilibrium is reached when the diffusion current exactly balances the drift current:
$$I_{\text{diff}} = I_{\text{drift}} \implies I_{\text{net}} = 0$$

Barrier Potential (V_0)

The potential difference developed across the depletion layer that halts further net diffusion of majority carriers is called the **Barrier Potential** (V_0).

$$V_0 \approx 0.7\ \text{V for Silicon}, \quad V_0 \approx 0.3\ \text{V for Germanium}$$

The corresponding internal electric field across the depletion layer of width $d \approx 10^{-6}$ m is extraordinarily strong:

$$E_i = \frac{V_0}{d} \approx \frac{0.7}{10^{-6}} = 7 \times 10^5\ \text{V/m}$$

Forward Bias Dynamics

- **Connection:** Positive terminal of battery connected to p-side; negative terminal to n-side.
- **Mechanism:** The applied external voltage V acts in opposition to the internal built-in barrier potential V_0.
- **Effective Barrier:** Lowers to $(V_0 - V)$.
- **Depletion Width:** Depletion layer width decreases significantly ($W \propto \sqrt{V_0 - V}$).
- **Conduction:** When $V > V_0$ (exceeding the **cut-in** or **knee voltage**), majority carriers easily surmount the reduced barrier. Diffusion current surges exponentially into the milliampere (mA) regime.

Reverse Bias Dynamics

- **Connection:** Positive terminal connected to n-side; negative terminal to p-side.
- **Mechanism:** The applied voltage V acts in the same direction as the internal built-in barrier potential V_0.
- **Effective Barrier:** Increases to $(V_0 + V)$.
- **Depletion Width:** Depletion layer widens ($W \propto \sqrt{V_0 + V}$).
- **Conduction:** Majority carrier diffusion is completely choked off. Only minority carriers thermally generated within or near the depletion layer are swept across by the strong field, producing a tiny, voltage-independent **Reverse Saturation Current** (I_s).
  - For Germanium: $I_s \sim$ microamperes (μA).
  - For Silicon: $I_s \sim$ nanoamperes (nA) (negligible).

Dynamic (AC) Resistance

Unlike an ohmic resistor whose I-V characteristic is linear, a semiconductor diode is a non-linear, non-ohmic device. Its resistance varies at each operating point:

- **Static (DC) Resistance:** $R_{\text{dc}} = \frac{V}{I}$
- **Dynamic (AC) Resistance:** The reciprocal of the slope of the I-V curve:
$$r_d = \frac{\Delta V}{\Delta I}$$

DYNAMIC RESISTANCE CALCULATION

Problem: In a forward-biased Silicon diode, when the applied voltage increases from 0.65 V to 0.70 V, the forward current increases from 10 mA to 35 mA. Determine the dynamic resistance of the diode in this region.

Solution:

$$\Delta V = 0.70 \text{ V} - 0.65 \text{ V} = 0.05 \text{ V} = 50 \text{ mV}$$
$$\Delta I = 35 \text{ mA} - 10 \text{ mA} = 25 \text{ mA} = 25 \times 10^{-3} \text{ A}$$
$$r_d = \frac{\Delta V}{\Delta I} = \frac{0.05 \text{ V}}{25 \times 10^{-3} \text{ A}} = 2.0 \ \ \Omega$$

Concept Module 4: Diode as a Rectifier (AC to DC Conversion)

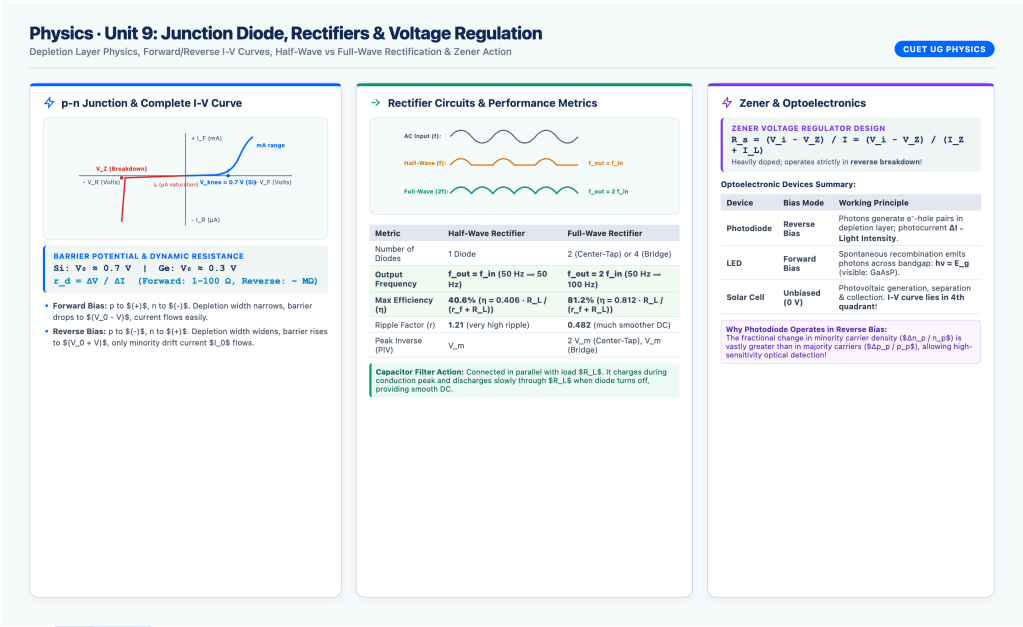


Figure 2: p-n junction I-V curves, half-wave & full-wave rectifier topologies with smoothing filter, and Zener diode voltage regulator circuit.

Rectification Principle

A p-n junction diode offers very low resistance ($r_f \approx 1$ to $10 \ \ \Omega$) when forward-biased and extremely high resistance ($r_r \approx 10^5$ to $10^7 \ \ \Omega$) when reverse-biased. This unilateral conduction property allows the diode to convert alternating current (bidirectional AC) into pulsating direct current (unidirectional DC).

Half-Wave Rectifier

In a half-wave rectifier, a single diode is connected in series with the secondary of a transformer and the load resistor R_L.

- **Positive Half-Cycle:** Diode anode is positive relative to cathode $\Longrightarrow$ Diode is forward-biased and conducts, producing a half-sine voltage drop across R_L.
- **Negative Half-Cycle:** Diode anode is negative relative to cathode $\Longrightarrow$ Diode is reverse-biased and blocks conduction ($I = 0$).
- **Output Frequency:** The output pulses once per full AC cycle:
$$f_{\text{out}} = f_{\text{in}}$$
- **Maximum Theoretical Efficiency (η):**
$$\eta = \frac{P_{\text{dc}}}{P_{\text{ac}}} = \frac{0.406 R_L}{r_f + R_L} \Longrightarrow \eta_{\max} = 40.6\%$$
- **Ripple Factor (r):** Measures residual AC fluctuations in the output:

$$r = \sqrt{\left(\frac{I_{\text{rms}}}{I_{\text{dc}}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1} \approx 1.21$$

Full-Wave Rectifiers

Full-wave rectifiers conduct on both halves of the AC cycle, producing continuous unidirectional pulses.

1. **Center-Tapped Full-Wave Rectifier:**
   - Utilizes a center-tapped transformer and two diodes (D_1 and D_2).
   - **Positive Half-Cycle:** D_1 is forward-biased and conducts; D_2 is reverse-biased (OFF). Current flows through R_L from top to bottom.
   - **Negative Half-Cycle:** D_2 is forward-biased and conducts; D_1 is reverse-biased (OFF). Current flows through R_L in the **exact same direction**.
   - **Peak Inverse Voltage (PIV):** Maximum reverse voltage across the non-conducting diode is $2V_m$.
2. **Bridge Full-Wave Rectifier:**
   - Employs 4 diodes in a bridge configuration without requiring an expensive center-tapped transformer.
   - Two diodes conduct in series on each half-cycle: D_1, D_3 on positive half; D_2, D_4 on negative half.
   - **PIV:** V_m across each non-conducting diode.

KEY FORMULA

Full-Wave Rectifier Key Metrics:

$$\text{Output Ripple Frequency:} \quad f_{\text{out}} = 2f_{\text{in}} \quad (\text{e.g., } 50 \text{ Hz AC} \rightarrow 100 \text{ Hz DC pulses})$$

$$\text{Maximum Efficiency:} \quad \eta = \frac{0.812 R_L}{r_f + R_L} \Longrightarrow \eta_{\max} = 81.2\%$$

$$\text{Ripple Factor:} \quad r = \sqrt{\left(\frac{I_{\text{rms}}}{I_{\text{dc}}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} \approx 0.48$$

Smoothing Filters (Capacitor Filter)

The rectified DC output is a pulsating voltage consisting of a steady DC component superimposed with high-frequency AC ripple harmonics.

- A capacitor of large capacitance C is connected in parallel across the load resistor R_L (shunt capacitor filter).
- **Mechanism:** As the rectifier voltage rises to peak V_m, the capacitor charges rapidly. When the rectifier voltage drops below V_m, the diode turns OFF, and the capacitor slowly discharges through R_L with time constant $\tau = R_L C$.
- If $R_L C \gg T$ (where $T = \frac{1}{f}$ is the AC time period), the capacitor discharges very little before the next voltage peak recharges it, yielding a nearly smooth DC output.

Concept Module 5: Special Purpose Diodes: Zener & Optoelectronics

Zener Diode & Voltage Regulation

A Zener diode is a heavily doped Silicon p-n junction diode engineered specifically to operate safely and continuously in the reverse breakdown regime without destruction.

- **Heavy Doping Effect:** Heavy doping (~ 1 in 10^4) makes the depletion region exceptionally thin ($W < 1$ μm).
- **Zener Breakdown Mechanism:** Even a modest reverse voltage ($V_Z \approx 5$ to 10 V) establishes an immense electric field ($E = \frac{V_Z}{W} > 10^6$ V/m) across the junction. This electric field is strong enough to

directly pull valence electrons out of covalent bonds (**field emission**), releasing an enormous burst of electron-hole pairs and causing an abrupt current rise at a precisely fixed breakdown voltage V_Z.

- **Avalanche Breakdown Comparison:** In lightly doped diodes with wide depletion layers, accelerated minority carriers collide with lattice atoms, knocking out secondary carriers (**impact ionization**), leading to avalanche multiplication ($V_{\text{BR}} > 6$ V).

Sizing the Zener Voltage Regulator Resistor (R_S)

A Zener diode acts as a constant-voltage shunt regulator:

- It is connected in reverse bias across the unregulated DC supply in series with a dropping resistor R_S.
- The load R_L is placed in parallel with the Zener diode, so $V_L = V_Z = \text{constant}$.
- Applying Kirchhoff's Current Law at the load node:

$$I_S = I_Z + I_L$$

$$V_{\text{in}} - V_Z = I_S R_S \Longrightarrow R_S = \frac{V_{\text{in}} - V_Z}{I_Z + I_L}$$

- **Fluctuation Compensation:** If V_{in} increases, the excess voltage is dropped across R_S by an increase in I_Z, while $V_L = V_Z$ remains locked. If load resistance R_L decreases (drawing more I_L), the Zener diode current I_Z decreases by the same amount, keeping total current I_S and output voltage V_Z invariant.

ZENER VOLTAGE REGULATOR CALCULATION

Problem: A 10 V Zener diode is used to maintain a constant voltage across a 1 k Ω load from an unregulated 15 V DC supply. If the minimum Zener current is 5 mA, calculate the required series resistance R_S and its power dissipation.
Solution:

$$V_L = V_Z = 10 \text{ V}$$

Load current: $I_L = \frac{V_L}{R_L} = \frac{10 \text{ V}}{1000 \ \Omega} = 10 \text{ mA} = 0.010 \text{ A}$ Total supply current: $I_S = I_Z + I_L = 5 \text{ mA} + 10 \text{ mA} = 15 \text{ mA} = 0.015 \text{ A}$ Voltage drop across series resistor: $V_{R_S} = V_{\text{in}} - V_Z = 15 \text{ V} - 10 \text{ V} = 5 \text{ V}$

$$R_S = \frac{V_{\text{in}} - V_Z}{I_S} = \frac{5 \text{ V}}{0.015 \text{ A}} \approx 333.3 \ \ \Omega$$

Power dissipated in R_S: $P_{R_S} = I_S^2 R_S = (0.015)^2 \times 333.3 = 0.075 \text{ W} = 75 \text{ mW}$.

Optoelectronic Junction Devices

1. **Photodiode:**
   - **Operation:** Operates under **reverse bias** with a transparent optical window.
   - **Principle:** When photons of energy $h\nu \geq E_g$ strike the depletion region, electron-hole pairs are photogenerated. The built-in electric field sweeps electrons to the n-side and holes to the p-side before recombination, increasing the reverse drift current.
   - **Why Reverse Bias?** In a forward-biased diode, the majority carrier current is large (mA), so the fractional change ($\Delta\frac{I}{I}$) due to illumination is unnoticeable. In reverse bias, the dark current is tiny (nA), so the fractional change ($\Delta\frac{I}{I_s}$) is dramatic and easily detected.
   - **Application:** High-speed optical communication, light meters, CD players.
2. **Light Emitting Diode (LED):**
   - **Operation:** Operates under **forward bias**.
   - **Principle:** When heavily forward-biased, majority electrons inject into the p-region and holes into the n-region. At the boundary, electrons and holes recombine radiatively, releasing energy as photons of frequency ν:

$$h\nu \approx E_g \Longrightarrow \lambda \approx \frac{hc}{E_g}$$

   - **Material Criterion:** Silicon and Germanium are indirect bandgap semiconductors where recombination releases phonons (heat). Direct bandgap compound semiconductors like Gallium Arsenide Phosphide (GaAsP) are required. For visible light emission (400 to 700 nm), the bandgap must satisfy $E_g \geq 1.8$ eV.
3. **Solar Cell:**

- **Operation:** Operates at **zero bias** (no external battery connected).
- **Principle:** Converts light energy directly into electrical energy via three sequential processes: (1) **Generation** of electron-hole pairs near junction by photons ($h\nu > E_g$); (2) **Separation** of electrons and holes by the built-in electric field; (3) **Collection** at the respective electrodes, creating a photo-EMF.
- **I-V Curve:** Plotted strictly in the **fourth quadrant** because the cell supplies power to the circuit rather than absorbing it.
- **Key Parameters:** Open-Circuit Voltage (V_{oc}) and Short-Circuit Current (I_{sc}).

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Concept Module 7: Conceptual Synthesis & Mindmap

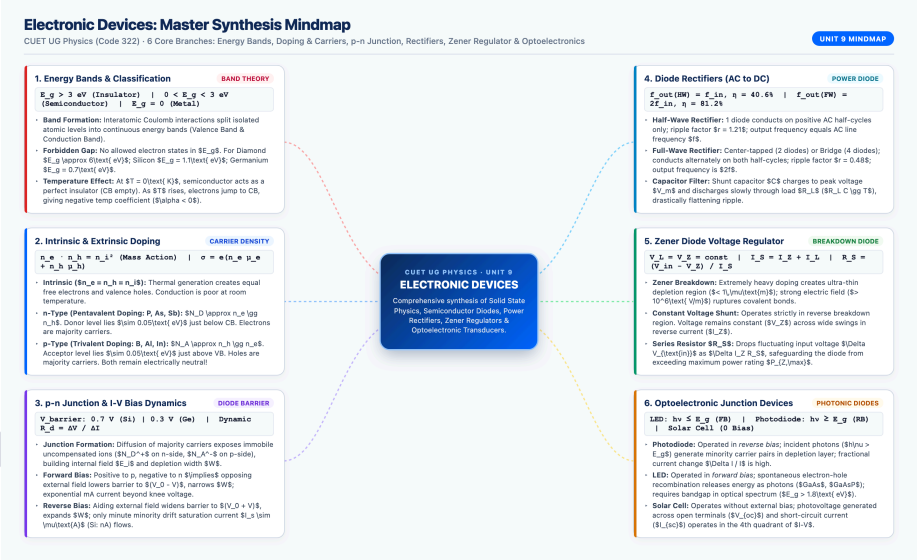


Figure 3: Electronic Devices Master Synthesis Mindmap covering solid-state physics, junction dynamics, rectifiers, and optoelectronics.

High-Yield Formula Cheatsheet & Key Traps

MASTER REVISION FORMULA MATRIX: ELECTRONIC DEVICES

- **Band Gap Criterion:** Insulator ($E_g > 3$ eV), Semiconductor ($0 < E_g < 3$ eV), Metal ($E_g = 0$)
- **Intrinsic Carrier Law:** $n_e = n_h = n_i, \quad \sigma = en_i(\mu_e + \mu_h)$
- **Mass-Action Law:** $n_e \cdot n_h = n_i^2$ (holds for intrinsic and extrinsic at equilibrium)
- **Barrier Potentials:** $V_0 \approx 0.7$ V (Si), $V_0 \approx 0.3$ V (Ge)
- **Depletion Width:** $W \propto \sqrt{V_0 - V}$ (Forward Bias); $W \propto \sqrt{V_0 + V}$ (Reverse Bias)
- **Dynamic Resistance:** $r_d = \frac{\Delta V}{\Delta I}$
- **Half-Wave Rectifier:** $f_{\text{out}} = f_{\text{in}}$, $\eta_{\max} = 40.6\%$, Ripple $r = 1.21$
- **Full-Wave Rectifier:** $f_{\text{out}} = 2f_{\text{in}}$, $\eta_{\max} = 81.2\%$, Ripple $r = 0.48$
- **Zener Voltage Regulator:** $V_L = V_Z, \quad I_S = I_Z + I_L, \quad R_S = \frac{V_{\text{in}} - V_Z}{I_Z + I_L}$
- **LED Wavelength Condition:** $\lambda = \frac{hc}{E_g} \leq \frac{1242 \text{ eV} \cdot \text{nm}}{E_g} \quad (E_g \geq 1.8 \text{ eV for visible})$

THE SEVEN FATAL CUET TRAPS IN ELECTRONIC DEVICES

1. **Neutrality of Doped Semiconductors:** Always remember that n-type and p-type semiconductors are **electrically neutral**. Donor/acceptor ions balance free carriers.
2. **Rectifier Output Frequency:** Half-wave: $f_{\text{out}} = f_{\text{in}}$. Full-wave: $f_{\text{out}} = 2f_{\text{in}}$.
3. **Photodiode vs LED Bias:** Photodiodes operate in **reverse bias**; LEDs operate in **forward bias**; Solar cells operate with **no external bias**.
4. **Ideal Diode Analysis Rules:** Forward-biased ideal diode is a closed switch ($V = 0, R = 0$); reverse-biased is an open switch ($I = 0, R = \infty$).
5. **Zener Diode Heavy Doping:** Zener diodes are **heavily doped**, producing an extremely thin depletion layer and high internal electric field ($\sim 10^6$ V/m).
6. **Photodiode Reverse Saturation Current Fractional Change:** Measured in reverse bias because the fractional increase in minority carrier reverse current ($\Delta\frac{I}{I_0}$) is vastly larger and easier to measure than in forward bias.
7. **Temperature Coefficient of Resistance:** Semiconductors have **negative temperature coefficient** ($\alpha < 0$): conductivity increases exponentially with temperature.

Practice Questions — Part 1 (Q1 to Q5)

Q1. Carbon, Silicon, and Germanium each have four valence electrons. Their forbidden energy gaps are represented by $(E_g)_{\text{C}}$, $(E_g)_{\text{Si}}$, and $(E_g)_{\text{Ge}}$ respectively. Which of the following relationships is correct?

(A) $(E_g)_{\text{C}} > (E_g)_{\text{Si}} > (E_g)_{\text{Ge}}$
(B) $(E_g)_{\text{C}} < (E_g)_{\text{Si}} < (E_g)_{\text{Ge}}$
(C) $(E_g)_{\text{C}} = (E_g)_{\text{Si}} = (E_g)_{\text{Ge}}$
(D) $(E_g)_{\text{Si}} > (E_g)_{\text{C}} > (E_g)_{\text{Ge}}$

Sol. Q1. Carbon (diamond) has $E_g \approx 5.4$ eV, Silicon has $E_g = 1.1$ eV, and Germanium has $E_g = 0.7$ eV. Therefore, $(E_g)_{\text{C}} > (E_g)_{\text{Si}} > (E_g)_{\text{Ge}}$.
Answer: (A)

Q2. As the temperature of an intrinsic semiconductor is increased from 0 K to 300 K, its electrical resistivity:

(A) Increases linearly
(B) Remains strictly constant
(C) Decreases exponentially
(D) First decreases and then increases

Sol. Q2. At 0 K, a semiconductor is an insulator. Increasing temperature thermally generates electron-hole pairs, reducing electrical resistivity ($\alpha < 0$).
Answer: (C)

Q3. To obtain a p-type semiconductor, an intrinsic Silicon crystal should be doped with trace amounts of which of the following elements?

(A) Phosphorus
(B) Arsenic
(C) Antimony
(D) Indium

Sol. Q3. A p-type semiconductor requires a trivalent acceptor dopant (Boron, Aluminium, Gallium, Indium). Phosphorus, Arsenic, and Antimony are pentavalent donors.
Answer: (D)

Q4. A sample of pure Silicon has an intrinsic carrier concentration of 1.5×10^{16} m^{-3}. It is doped with Phosphorus atoms to a donor concentration of $N_D = 4.5 \times 10^{21}$ m^{-3}. The resulting hole concentration in the doped semiconductor is:

(A) 5.0×10^{10} m^{-3}
(B) 3.0×10^{10} m^{-3}
(C) 5.0×10^{8} m^{-3}
(D) 2.0×10^{15} m^{-3}

Sol. Q4. Here $n_e \approx N_D = 4.5 \times 10^{21}$ m^{-3}. By the Mass-Action Law: $n_h = \frac{n_i^2}{n_e} = \frac{(1.5\times10^{16})^2}{4.5\times10^{21}} = 5.0 \times 10^{10}$ m^{-3}.
Answer: (A)

Q5. When a p-n junction diode is forward-biased, the width of the depletion layer and the height of the potential barrier respectively:

(A) Decrease, Decrease
(B) Increase, Increase
(C) Increase, Decrease
(D) Decrease, Increase

Sol. Q5. Forward bias opposes the built-in field, reducing effective barrier height to $(V_0 - V)$ and narrowing the depletion layer width.
Answer: (A)

Practice Questions — Part 2 (Q6 to Q10)

Q6. The barrier potential of a Silicon p-n junction is 0.7 V and the width of the depletion layer is 1.0×10^{-6} m. The magnitude of the built-in electric field existing across the junction is:

(A) 7.0×10^{5} V/m **(B)** 7.0×10^{3} V/m
(C) 1.4×10^{6} V/m **(D)** 3.5×10^{5} V/m

Sol. Q6. The electric field across the depletion region is: $E = \frac{V_0}{d} = \frac{0.7\text{ V}}{1.0\times10^{-6}\text{ m}} = 7.0 \times 10^{5}$ V/m.
Answer: (A)

Q7. In the forward characteristic of a semiconductor diode, the current changes from 15 mA to 30 mA when the forward voltage is changed from 0.70 V to 0.73 V. The dynamic forward resistance of the diode is:

(A) 1.0 Ω **(B)** 2.0 Ω
(C) 4.0 Ω **(D)** 0.5 Ω

Sol. Q7. Dynamic resistance: $r_d = \frac{\Delta V}{\Delta I} = \frac{0.73\text{ V} - 0.70\text{ V}}{30\times10^{-3}\text{ A} - 15\times10^{-3}\text{ A}} = \frac{0.03\text{ V}}{15\times10^{-3}\text{ A}} = 2.0\ \Omega$.
Answer: (B)

Q8. An alternating voltage of frequency 50 Hz is applied to a full-wave rectifier. The fundamental frequency of the ripple in the rectified output voltage is:

(A) 25 Hz **(B)** 50 Hz
(C) 200 Hz **(D)** 100 Hz

Sol. Q8. In a full-wave rectifier, both half-cycles of the AC input are rectified, doubling output ripple frequency: $f_{\text{out}} = 2f_{\text{in}} = 2 \times 50\text{ Hz} = 100\text{ Hz}$.
Answer: (D)

Q9. The maximum theoretical power rectification efficiency of an ideal full-wave rectifier circuit without filter is:

(A) 40.6% **(B)** 50.0%
(C) 81.2% **(D)** 100.0%

Sol. Q9. The maximum theoretical efficiency of a half-wave rectifier is $\eta_{\max} = 40.6\%$; for a full-wave rectifier it is exactly twice: $\eta_{\max} = 81.2\%$.
Answer: (C)

Q10. Which of the following statements regarding a Zener diode is completely correct?

(A) It is lightly doped and operates in the forward-bias region **(B)** It is heavily doped and operates in the reverse breakdown region
(C) It has a very wide depletion region of several millimetres **(D)** It is destroyed immediately upon reaching the breakdown voltage

Sol. Q10. A Zener diode is heavily doped, creating an ultra-thin depletion layer (< 1 μm). It operates stably in reverse breakdown where voltage remains constant.
Answer: (B)

Practice Questions — Part 3 (Q11 to Q15)

Q11. A Zener diode having a breakdown voltage of 12 V is connected in a voltage regulator circuit with a series resistance $R_S = 200\ \Omega$ across an unregulated input DC voltage of 20 V. If the load resistance is $R_L = 1$ k Ω, the current flowing through the Zener diode is:

(A) 40 mA **(B)** 12 mA
(C) 28 mA **(D)** 52 mA

Sol. Q11. $V_{R_S} = 20\text{ V} - 12\text{ V} = 8\text{ V} \Longrightarrow I_S = \frac{8}{200} = 40$ mA. Load current $I_L = \frac{12}{1000} = 12$ mA. Zener diode current $I_Z = I_S - I_L = 40\text{ mA} - 12\text{ mA} = 28$ mA.
Answer: (C)

Q12. A photodiode is operated under reverse bias rather than forward bias because:

(A) Reverse bias increases the barrier potential to infinity **(B)** The fractional change in minority carrier reverse current upon illumination is far easier to measure
(C) Photons can only generate carriers in a reverse-biased junction **(D)** Forward bias permanently burns the optical window

Sol. Q12. In reverse bias, dark current is minute (minority carriers). Photogenerated carriers create a large fractional change ($\Delta\frac{I}{I}$), which is easily measured.
Answer: (B)

Q13. To emit visible light in the wavelength range of 400 nm to 700 nm, the semiconductor material used to fabricate a Light Emitting Diode (LED) must have a bandgap energy E_g of at least:

(A) 0.7 eV **(B)** 1.1 eV
(C) 1.8 eV **(D)** 3.2 eV

Sol. Q13. For visible emission: $E_g \geq \frac{hc}{\lambda_{\max}} = \frac{1242\text{ eV}\cdot\text{nm}}{700\text{ nm}} \approx 1.8$ eV. Silicon (1.1 eV) and Germanium (0.7 eV) emit in the infrared.
Answer: (C)

Q14. The current-voltage ($I - V$) characteristic of a solar cell is plotted in which quadrant of the Cartesian coordinate system?

(A) First quadrant **(B)** Second quadrant
(C) Third quadrant **(D)** Fourth quadrant

Sol. Q14. A solar cell operates without external bias and supplies power to an external load ($V > 0, I < 0$). Thus, its characteristic is plotted in the fourth quadrant.
Answer: (D)

Q15. In the circuit shown, two ideal diodes D_1 and D_2 are connected in series with resistors $R_1 = 100\ \Omega$ and $R_2 = 200\ \Omega$ across a 10 V DC source. Diode D_1 is forward-biased and D_2 is reverse-biased in their respective parallel branches. The total current drawn from the 10 V source is:

(A) 0.15 A **(B)** 0.10 A
(C) 0.05 A **(D)** 0 A

Sol. Q15. Forward-biased D_1 acts as a short circuit ($I_1 = \frac{10}{100} = 0.10$ A). Reverse-biased D_2 acts as an open circuit ($I_2 = 0$). Total current $I = 0.10\text{ A} + 0 = 0.10$ A.
Answer: (B)